# SDC Hackathon 2026 — Phase 1: Preliminary Project Proposal

**Hosted by Xiamen University Malaysia (XMUM) AI Club · Submission Deadline: 25 October 2026**

*Instructions (from the official template): Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.*

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# 1. Team and Project Overview

| **Field** | **Response** |
|---|---|
| Team Name | Hoo Lee Sheet |
| Selected Domain | Domain 1: The Aiversecity Future-Ready Workforce Domain |
| Selected SDG Track (1–17) | Primary: SDG 8 (Decent Work and Economic Growth): Laluan helps underemployed and AI-exposed workers move towards decent, more productive jobs (target 8.5). Secondary: SDG 4 (Quality Education): it builds pathways from free and subsidised courses so adults gain skills relevant for employment (target 4.4). Secondary: SDG 10 (Reduced Inequalities): Laluan needs no login, puts free and subsidised courses before paid ones, and lets users prefer courses taught in English, Bahasa Malaysia or Chinese, so workers with less access to paid career advice or English-only courses can still get a sourced pathway (target 10.2). |
| Project / Solution Name | Laluan |

| Field | Response |
|---|---|
| Name (Leader) | Tan Bing Jie |
| Email & Phone (Leader) | Email: tanbingjie99@gmail.com<br>Phone: 0128002983 |
| University Affiliation(s) | Xiamen University Malaysia (XMUM) |
| Number of Team Members | 3 Members:<br>● Tan Bing Jie (Leader)<br>● Henry Wong Yi Siang<br>● Tiffany Yek Sii Chee |

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# 2. Executive Summary & Problem Statement

## 2.1. The Problem Context

*Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved. [Maximum 250 words]*

Malaysian workers know AI is changing work, but most cannot answer three questions for themselves: how much does AI touch my job, which role should I move towards, and what is the cheapest credible route there? The answers are scattered across static reports, a national job portal and dozens of agency and course websites, and they date quickly as AI moves.

The need is large. In Malaysia, 56% of professionals use AI tools at work but only 26% have received training (2025 Hays Asia Salary Guide), and MOSTI says more than 600,000 workers must reskill within three to five years. The World Bank and ISIS Malaysia (2025) estimate 4.2 million workers (28%) are highly exposed to generative AI, more so among women, workers aged 25–34 and clerical workers. Meanwhile, 66.0% of graduates aged 24 or under work below their qualification, and underemployment is 47.1% for diploma holders versus 21.0% for degree holders (DOSM, 2024).

We focus on one slice: underemployed diploma holders and clerical workers aged 20–34, many of them women, at the moment they see routine tasks being automated and must choose what to learn next with little time and no budget. Existing tools cover only part of this: MYFutureJobs matches today's vacancies, MyMahir gives a one-off outlook, e-LATiH is a catalogue, and

LinkedIn or Coursera are backward-looking and mostly paid. So people spend scarce hours on the wrong courses and miss free or subsidised programmes.

## 2.2. The Proposed AI Solution

*Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage. [Maximum 300 words]*

Laluan (Malay for "pathway") is a web app that takes a user from "who I am today" to "where I can be in the AI era", with no login.

1. Skills profile intake: the user uploads a CV (synthetic samples in the demo) and answers a fixed 5-question interview. AI extracts roles, tasks, skills and education into a fixed schema; every skill is editable.
2. Future Skills Radar: an arXiv fetch script (run manually) counts papers for about 25 AI-skill keywords and stores dated snapshots. The radar shows rising and declining skills with source and date; a "30 days ago / today" switch shows recommendations changing as the data changes.
3. Explainable matching: synonym matches, then Gemini embeddings, map the user's skills to ESCO skills; deterministic code scores 40 ESCO roles (fit, AI exposure, augmentation) and lists the top missing skills with source chips.
4. Scenarios: three of WEF's four futures (Co-Pilot Economy, Age of Displacement, Supercharged Progress) re-rank the roles; a role in the top three in every future is marked "robust". Every card states "All scores and forecasts are estimates."
5. SDG mapping: every role and skill links to at least one SDG target with a confidence tier, and the user's impact interest nudges the ranking, so SDGs shape the choice.
6. Pathway: the user picks a role; deterministic code orders items from our curated course and programme list (free, then subsidised, then HRD Corp-claimable), with durations, boosting the preferred course language (English, Bahasa Malaysia or Chinese).

AI only parses CVs and job titles; code handles scores, explanations and pathways, so Laluan cannot invent jobs, courses or links. It supports career decisions and never makes hiring decisions.

Workshop skills applied: GenAI APIs (Claude Haiku 4.5 and Gemini structured output, Gemini embeddings), database architecture (Supabase Postgres, pgvector, row-level security) and cloud hosting (Replit, Vercel backup).

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# 3. Technical Architecture & Implementation Plan

## 3.1. Conceptual Tech Stack

*List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.*

| **Layer** | **Technology** | **What it does in Laluan** |
|---|---|---|
| Frontend | Next.js (App Router, TypeScript), Tailwind CSS, shadcn/ui | Screens: Landing (consent) → Profile → Interview → Results → Pathway, plus a Future Skills Radar page. UI in English only; an EN / BM / 中文 pill sets the preferred course language. Mobile-friendly. |
| Backend | Next.js API routes (server-side only) | Profile parsing, interview turns, matching and scoring, pathway building, latest signals. Keys never reach the browser. |
| GenAI | Vercel AI SDK with Zod-validated structured output. Chat model via getModel(): Anthropic Claude Haiku on Replit; Google Gemini gemini-3.8-flash on Vercel and as fallback if Haiku fails. Embeddings: gemini-embedding-001 (768 dimensions). | Chat model: CV upload and parsing, and occupation matching (/api/exposure-estimate) only when exact-name and alias matches fail; if nothing is close, the app shows "No close match". Embeddings: CV skills to ESCO skills (skill-match.ts) only when exact and synonym matches fail; not used for occupations. The 5-question interview is fixed; results, explanations and pathway ordering come |

| Layer | Technology | What it does in Laluan |
|---|---|---|
| | | from code and bundled data, with no model call. No model produces a score or a URL. |
| Scoring engine | TypeScript pure functions (lib/scoring.ts), unit-tested with Vitest | AI exposure, augmentation, skill momentum, role fit, scenario ranking, "robust" flag and the top-5 skills gap. Same input always gives the same output. |
| Database | Supabase (Postgres) with pgvector (HNSW index) and row-level security | Occupations, skills, tasks, dated signals, scenarios, SDG links, courses, programmes, profiles, results and pathways. No login; no CV file stored. |
| AI-trend signal | arXiv API, fetched by a script (scripts/fetch-arxiv.ts) run manually; the in-app Refresh button re-reads the stored snapshot | Paper counts per AI-skill keyword (last 90 days vs the 90 days before), stored as dated snapshots for the Future Skills Radar and the "30 days ago / today" switch. |
| Reference data | ESCO (40 occupations across 5 clusters, with skills), ILO GenAI exposure index (2025), Anthropic Economic Index, WEF Four Futures scenarios (3 of 4 used), curated free / subsidised / HRD Corp-claimable courses and government programmes | Every figure is stored with its source and date and shown as a source chip. Course rows carry language (en / ms / zh), hours, cost type and a last-checked date. |
| Hosting | Replit Autoscale deployment (main); Vercel (backup) | Live prototype for judges; Vercel only if Replit has problems during judging. |
| Source control | GitHub (Jie-coder/hoo-lee-sheet-sdc 2026), Apache License 2.0 | Pull-request workflow; every pull request reviewed by a |

| Layer | Technology | What it does in Laluan |
|---|---|---|
| | | human before merging (no CI pipeline). |
| Optional | RunPod: not used | Not used. The committee confirmed RunPod is only required for GPU work; Laluan calls hosted model APIs (Claude Haiku, Gemini) and runs no self-hosted GPU workloads. |

### System block diagram

```
User's browser (no login)

  Landing/consent -> Profile (CV + edit skills) -> Interview (5 Qs)

  -> Results -> Pathway;  Future Skills Radar page

        |

        v  HTTPS

Replit Autoscale (main)  /  Vercel (backup)

  Next.js app + server-only API routes

        |-- getModel(): Claude Haiku (Replit) / gemini-3.8-flash (Vercel,
        |     fallback): CV parsing; closest-occupation pick

        |-- Fixed 5-question interview (no Gemini)

        |-- /api/exposure-estimate: exact name -> alias -> AI pick
        |     -> "No close match"
        |-- skill-match.ts: exact -> synonym -> gemini-embedding-001
        |     + pgvector (CV skills to ESCO skills only)

        |-- lib/scoring.ts: fit, exposure, augmentation, 3 scenarios,

        |                   robust flag, skills gap, explanations,
        |                   pathway ordering (deterministic, no Gemini)

        `-- Supabase Postgres + pgvector

              occupations, skills, signals (dated), scenarios,

              sdg_links, courses, programs, results, pathways

                 ^                              ^
```

```
arXiv fetch script (manual):          Data import script:

arXiv API -> keyword counts ->        ESCO, ILO, Anthropic Economic Index,

dated signal snapshots                curated courses/programmes CSV
```

(No self-hosted GPU workloads: all model calls go to hosted APIs (Claude Haiku, Gemini); RunPod not used)

**How each Domain 1 must-have is delivered**

| **Must-have** | **How Laluan delivers it** |
|---|---|
| 1. Skills profile intake | CV upload (PDF) parsed by an LLM (Claude Haiku on Replit; Gemini on Vercel and as fallback) into roles, tasks, skills and education, plus a 5-question slot-filling interview (time share per task, tools and AI use, hours per week and budget, work status, impact interest). The parsed skill list is editable. Demo uses three synthetic CVs; uploads are never stored. No citizenship question. |
| 2. AI-trend signal layer | arXiv paper counts for about 25 AI-skill keywords, refreshed by a manually run fetch script, each stored with its date and source link. The Future Skills Radar shows rising and declining skills and roles; the "30 days ago / today" switch visibly changes the output. |
| 3. Future-role matching | 40 ESCO roles ranked by an explainable score: 0.5 × fit + 0.3 × scenario demand + 0.2 × (1 − exposure penalty), plus a small SDG-interest bonus. Each role card shows fit %, the top 5 missing skills and source chips; the plain-language explanation is built by code from the same data, with no Gemini call. |
| 4. Personalised reskilling pathway | Ordered, week-by-week plan with estimated hours, built only from our curated catalogue, ordered free → subsidised → HRD Corp-claimable → paid, filtered by hours and budget and boosted for the preferred course language (EN / BM / Chinese). Programmes |

| Must-have | How Laluan delivers it |
|---|---|
| | that need citizenship or other unknown criteria show "please check eligibility yourself". |
| 5. SDG mapping | Every role and every skill links to at least one SDG target (e.g. 4.3, 4.4, 8.2, 8.5) with a tier (direct / supporting / inferred), a reason and a source. The pathway shows which targets the user's chosen move advances. |
| 6. Handling uncertainty | Three of the four scenarios in WEF's "Four Futures for Jobs in the New Economy: AI and Talent in 2030" (Jan 2026): Co-Pilot Economy, Age of Displacement and Supercharged Progress (we do not use the fourth, Stalled Progress), implemented as a labelled, illustrative parameter table; roles re-rank per scenario and a "robust" badge marks roles that stay in the top three in all of them. Every score card states "All scores and forecasts are estimates"; every results screen states "Laluan supports your decision; it does not make hiring decisions." |

**Hardest risk and fallback.** Our hardest risks are (a) keeping the trend data reliable and updatable and (b) the AI inventing jobs, courses or numbers. Mitigations: signals are stored as dated snapshots in Supabase, so the radar still works if arXiv is unreachable; all numbers come from code; the model only parses CVs and, when exact and alias matches fail, picks the closest listed occupation; it never picks courses, so it cannot write a link. If the model API, the network or a free tier fails during judging, a DEMO_MODE cache of the three synthetic personas keeps the demo running, and Vercel is the backup host. Only public or synthetic data is used.

Live prototype link: https://hoo-lee-sheet-sdc-2026.replit.app (backup: https://laluan.vercel.app). Repository: https://github.com/Jie-coder/hoo-lee-sheet-sdc2026

*Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.*

**Yes. Replit** (sponsor-provided Replit Core) is our main development and hosting platform: the GitHub repo is imported into the team leader's Replit account, API keys are kept in Replit Secrets, and the live prototype is published as a Replit Autoscale deployment. Proof of use will

be included in the Technical & Tools Declaration. **RunPod (sponsor-provided credit): not used. The committee confirmed RunPod is only required for GPU work, and Laluan uses hosted model APIs (Claude Haiku, Gemini) with no self-hosted GPU workloads.** Our GenAI API, Google Gemini, is used through a free Google AI Studio key and is not sponsor-provided. Claude Haiku (claude-haiku-4-5) runs on Replit through Replit's built-in AI Integrations (Anthropic), with no separate Anthropic key; usage is billed to the organiser-provided Replit Core credits (handbook 9.1.1), so this is a sponsor-provided resource.

## 3.2. Design-Thinking & Deployment Strategy

*Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2. [Maximum 200 words]*

Empathise: before choosing an idea we reviewed Malaysian labour data and existing tools (MYFutureJobs, MyMahir, e-LATiH) and built a persona: Aisyah, 23, a diploma-holding admin clerk in Penang with six study hours a week and no budget.

Define: we framed her need as three questions (how much AI touches my job, where to move, the cheapest credible route) and mapped each to a Domain 1 must-have.

Ideate: options and trade-offs are recorded in our decisions log, for example 40 checked roles before 150, five interview questions instead of eight, no login and no citizenship question.

Prototype: wireframes for five screens, a results page on mock data, three synthetic CVs and a unit-tested scoring engine.

Test: we walk each synthetic persona through the flow and check that every number shows a dated source.

Deploy: Replit Autoscale is our main host with Vercel as backup; keys are stored as environment secrets on each host (Replit Secrets for the primary deployment, Vercel environment variables for the backup) and no key values appear anywhere; a DEMO_MODE cache of persona results keeps the demo working if an API or the network fails; features were frozen on 23 October; final polish ran through 24 October.

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# 4. Preliminary Environmental & SDG Impact Framework

*To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used. The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

| **Metric Category** | **Your Response** |
|---|---|
| **Target SDG Sub-Target(s)** — Identify the exact UN SDG sub-target your project addresses | Primary — Target 8.5: "By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value." Secondary — Target 4.4: "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Secondary — Target 10.2: "By 2030, empower and promote the social, economic and political inclusion of all, irrespective of age, sex, disability, race, ethnicity, origin, religion or economic or other status." [1] |
| **Current Problem Baseline** — What is the starting condition of the problem, expressed as a number? | • 47.1% of diploma holders in Malaysia are in skill-related underemployment (21.0% of degree holders) [3]. • 66.0% of graduates aged 24 or under work below their qualification (32.2% of all graduates; 1.60 million people) [3]. • 4.2 million Malaysian workers (28%) are highly exposed to generative AI [2]. • 56% of Malaysian professionals use AI tools at work, but only 26% have received training [5]. • Product baseline: none of the existing tools we |

| Metric Category | Your Response |
|---|---|
| | reviewed (MYFutureJobs, MyMahir, e-LATiH, LinkedIn / Coursera, Lightcast / Eightfold) combines all six Domain 1 must-haves (team review, 19 Oct 2026). |
| **Core Measurable KPI** — The specific unit you will track | **Primary KPI: pathway gap coverage** — the % of a user's top-5 missing skills (for the role they choose) that their generated pathway closes using only free, subsidised or HRD Corp-claimable items. **Trust KPIs:** (a) % of on-screen scores, roles, courses and SDG links that show a dated source; (b) % of the 40 roles and their skills tagged to at least one SDG target; (c) minutes from start to a finished pathway; (d) agreement rate when two members independently hand-check 50 role–SDG pairs, with Cohen's kappa. |
| **Phase 2 Impact Target** — Your realistic, percentage-based target by the final prototype presentation | • Pathway gap coverage of at least 80% for each of the three synthetic personas, and a mean of at least 90% across them (measure: share of each persona's top-5 missing skills closed by at least one free, subsidised or HRD Corp-claimable pathway step). • 100% of recommendations show a dated source (design rule: every recommendation cites a source). • 100% of the 40 roles and their skills carry at least one SDG target (design rule). • Each persona reaches a finished pathway in under 3 minutes. • SDG link agreement of at least 80% overall on the 50 hand-checked role–SDG pairs and at least 90% for direct-tier links, with two members labelling independently; Cohen's kappa also reported, target at least 0.6. |

**Sources cited**


1. United Nations, Sustainable Development Goals: Goal 8, Goal 4 and Goal 10 targets (target wording checked against the official UN pages on 24 Oct 2026).

https://sdgs.un.org/goals/goal8; https://sdgs.un.org/goals/goal4; https://sdgs.un.org/goals/goal10
2. Cheng, Chong, Dornan & Farha Jasmin (July 2025), "Novel AI technologies and the future of work in Malaysia", policy brief, World Bank and ISIS Malaysia (4.2 million workers, 28% of the labour force, highly exposed). https://documents1.worldbank.org/curated/en/099092325013010451/pdf/P181093-2e5b89c5-f3be-43b3-868c-8890b74bef21.pdf
3. Department of Statistics Malaysia (DOSM), Graduates Statistics 2024 (skill-related underemployment: 66.0% of graduates aged 24 or under; 47.10% of diploma holders; 21.00% of degree holders). https://www.dosm.gov.my/uploads/release-content/file_20260401145140.pdf
4. World Economic Forum (January 2026), "Four Futures for Jobs in the New Economy: AI and Talent in 2030". Source of the three scenarios we use (Co-Pilot Economy, Age of Displacement, Supercharged Progress); the fourth, Stalled Progress, is not used. https://reports.weforum.org/docs/WEF_Four_Futures_for_Jobs_in_the_New_Economy_AI_and_Talent_in_2030_2025.pdf
5. Hays (2025), 2025 Hays Asia Salary Guide, Malaysia: 56% of professionals use AI tools at work; 26% have received training. https://www.hays.com.my/press-release/content/future-of-work-malaysia-technology-ai-impact
6. Malay Mail (7 January 2025), MOSTI Minister Chang Lih Kang: more than 600,000 workers must reskill within three to five years. https://www.malaymail.com/news/malaysia/2025/01/07/mosti-ai-disruption-looms-in-malaysia-600000-workers-must-reskill-in-three-to-five-years/162368
7. Gmyrek, P. et al. (2025), Generative AI and Jobs: A Refined Global Index of Occupational Exposure, ILO Working Paper 140. doi:10.54394/HETP0387. https://www.ilo.org/publications/generative-ai-and-jobs-refined-global-index-occupational-exposure
8. World Economic Forum, Future of Jobs Report 2025 (7 Jan 2025). https://www.weforum.org/publications/the-future-of-jobs-report-2025/ Used only for job figures (170 million jobs created and 92 million displaced by 2030; 39% of workers' core skills expected to change); the scenarios come from [4].
9. arXiv API (Cornell University), used for the trend signal. https://arxiv.org/help/api/index

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# 5. Eligibility & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

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# 6. IP & Originality Declaration

## 6.1. Pre-Existing / Prior-Owned Work

*Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?*

☑ Yes ☐ No

Our competition repository starts from a generic starter that Tan Bing Jie created before the hackathon (private repo Jie-coder/sdc-starter, tag starter-v0, commit 57171fe). It contains only reusable infrastructure (Next.js scaffold, UI components, an AI call wrapper with a mock mode, a Supabase connection and a status page) and no domain-specific features or content. It was built with AI coding assistance (Grok, using Claude Code) under human review. It was declared to the organisers under Rule 8.2 before the event (Discord ticket #ticket-0007, 17 Oct 2026), is imported as the first commit of the competition repo, and is documented in DECLARATION.md. All later work was created during the hacking period.

## 6.2. Third-Party Code, Datasets, or Models

*Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?*

☑ Yes ☐ No

| **Name** | **Type (code/dataset/model)** | **License** |
|---|---|---|
| Next.js | Code (framework) | MIT |
| React / React DOM | Code (library) | MIT |
| Tailwind CSS | Code (framework) | MIT |
| shadcn/ui | Code (UI components) | MIT |

| Name | Type (code/dataset/model) | License |
|---|---|---|
| Base UI (@base-ui/react) | Code (UI primitives) | MIT |
| Vercel AI SDK (ai, @ai-sdk/google) | Code (library) | Apache-2.0 |
| Zod | Code (validation library) | MIT |
| Supabase JS client | Code (library) | MIT |
| pgvector | Code (Postgres extension) | PostgreSQL License |
| lucide-react | Code (icons) | ISC |
| sonner, next-themes, tw-animate-css | Code (UI utilities) | MIT |
| class-variance-authority | Code (UI utility) | Apache-2.0 |
| Vitest | Code (testing) | MIT |
| Inter and Newsreader fonts | Code (fonts) | SIL Open Font License 1.1 |
| ESCO v1.2 occupations and skills (European Commission) | Dataset | Reuse free of charge with acknowledgement (Commission Decision 2011/833/EU); app must state "This service uses the ESCO classification of the European Commission". https://esco.ec.europa.eu/en/about-esco/faq |
| ILO GenAI exposure index, Working Paper 140 (2025) | Dataset | CC BY 4.0 (source: https://www.ilo.org/sites/default/files/2025-05/WP140_web.pdf, p.2) |
| Anthropic Economic Index (Hugging Face) | Dataset | Dataset card: "Data released under CC-BY, code released under MIT License" (no CC-BY version number stated on the card; the repository licence tag reads |

| Name | Type (code/dataset/model) | License |
|---|---|---|
| | | "mit"); release 2026-06-26. https://huggingface.co/datasets/Anthropic/EconomicIndex |
| arXiv API metadata | Dataset | CC0 (metadata) |
| WEF "Four Futures for Jobs in the New Economy: AI and Talent in 2030" (scenarios) and Future of Jobs Report 2025 (job figures) | Dataset (published report) | Four Futures: the report prints "© 2026 World Economic Forum. All rights reserved. No part of this publication may be reproduced or transmitted in any form or by any means…" (only scenario names and directions cited, no figures copied). WEF's general licence terms (CC BY-NC-ND 4.0 "unless otherwise indicated", https://www.weforum.org/about/licence-terms-on-the-use-of-forum-publications-and-materials/) do not override the report's own notice. https://reports.weforum.org/docs/WEF_Four_Futures_for_Jobs_in_the_New_Economy_AI_and_Talent_in_2030_2025.pdf; Future of Jobs Report 2025: All rights reserved (as printed in the report; we cite single figures with page numbers) |
| Curated course and programme catalogue (public provider pages, e.g. e-LATiH / HRD Corp) | Dataset (facts and links collected by the team; summaries in our own words) | No open licence; each provider's terms apply. We store only titles, links and own-words summaries. Coursera (94 rows): https://www.coursera.org/about/terms; Alison (36): |

| Name | Type (code/dataset/model) | License |
|---|---|---|
| | | https://alison.com/about/terms-of-use; e-LATiH (22): https://elatih.hrdcorp.gov.my/pages/terms-of-use; Microsoft Learn (18): https://learn.microsoft.com/en-us/legal/termsofuse; Cisco Networking Academy: Terms and Conditions for Use of Cisco Networking Academy Websites and Services, https://www.netacad.com/resources/legal/terms-conditions; OpenLearn: CC BY-NC-SA 4.0 unless otherwise stated, https://about.open.ac.uk/strategy-and-policies/policies-and-statements/copyright-openlearn; HRD Corp website: Licence not stated; used for citation/reference only (source: https://hrdcorp.gov.my/); MyFutureJobs: Licence not stated; used for citation/reference only (source: https://myfuturejobs.gov.my/); 11 single-course providers: not checked individually, each provider's own site terms apply |
| Three synthetic sample CVs | Dataset (created by the team with AI assistance) | Apache-2.0 (part of our repo) |
| Google Gemini, gemini-3.8-flash (chat model on Vercel and fallback, via getModel()); Anthropic Claude Haiku, | Model (pre-trained LLM, via API) | Google Gemini API Terms of Service (https://ai.google.dev/gemini-api/terms) |

| Name | Type (code/dataset/model) | License |
|---|---|---|
| claude-haiku-4-5 (chat model on Replit, via getModel(); via Replit AI Integrations; Replit and Anthropic terms apply) | | |
| Google gemini-embedding-001 (768 dimensions) | Model (embedding, via API) | Google Gemini API Terms of Service (https://ai.google.dev/gemini-api/terms) |

Services used: Replit, Supabase, Vercel and GitHub (free tiers). AI tools used during development are declared in full in the Technical & Tools Declaration (see "AI in development" below).
Other external data resources: OpenDOSM Salaries & Wages by Occupation (wages_occupation), CC BY 4.0, https://open.dosm.gov.my/data-catalogue/wages_occupation; DOSM salary survey major-group averages (salary fallback): Government Open Data Terms of Use 1.0 (DOSM open data portal), source: https://www.dosm.gov.my/portal-main/article/term-of-use; attribution: "Source: Department of Statistics Malaysia"; JobStreet Malaysia salary pages: the team manually checked salary figures on the JobStreet platform as a reference for the salary card (no scraping, no API); JobStreet terms allow personal, non-commercial use only and prohibit scraping and reproducing content without permission, https://my.jobstreet.com/terms.
AI in the product: Claude Haiku 4.5 (claude-haiku-4-5) is the main model. It runs on Replit through Replit AI Integrations, billed to the organiser-provided Replit Core credits. Google Gemini (gemini-3.8-flash) is the fallback on the Vercel deployment, using the team's own API key. gemini-embedding-001 is used for skill matching (CV skills to ESCO skills). The chat model only parses CVs and, when exact and alias matches fail, picks the closest listed occupation. AI exposure scores come straight from ILO Working Paper 140 occupation averages; no model produces a score. The three sample CVs are pre-parsed, so the demo does not depend on a live model call.
AI in development: the team's AI coding assistants are referred to as Grok. Grok wrote and reviewed code using Claude Code, Cursor cloud agents and, from 23 Oct, OpenCode with the DeepSeek model (light edits such as i18n strings and the first draft of the print receipt). Some work was hand-written, as recorded in the team's log (e.g. print CSS, demo PDF deletion and parts of PRs #56 to #58). Every pull request was reviewed by a human before merging. The team also used Google Stitch (Google Labs AI UI design tool, https://stitch.withgoogle.com/) for UI/visual inspiration for the website design only; no generated code is used in the product (minor).

## 6.3. Employment-Related IP Conflicts

*Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?*

☐ Yes ☑ No (all three team members are students; none is bound by such an agreement)

If yes, please describe the nature of the agreement and which member it concerns: Not applicable.

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# 7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



**Signature:** **Date:** 24/10/2026